



Suite 18.05,  
Level 18 Centrepoint North Mid Valley City,  
Lingkaran Syed Putra,  
59200 Kuala Lumpur W.P. Kuala Lumpur, Malaysia.

# Certificate of Analysis

## PRODUCT DETAILS

Batch

Date

Cell Source and Passage

Human Umbilical Cord (Wharton's Jelly), Passage 3

Storage Temperature

4 – 8°C

Viable Cell Count

| TEST                           | SPECIFICATION                                                                                                           | TEST RESULTS                | REMARKS                         |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------------------------------|
| Infectious Diseases Screening  | HBsAg; HCV Ab; HIV1/2 Ag/Ab; CMV Ab; RPR                                                                                | Not Detected / Non-Reactive | <b>QUALITY CONTROL APPROVED</b> |
| Morphology of Cells            | Spindle-like cells                                                                                                      | Small spindle shaped cells  | <b>QUALITY CONTROL APPROVED</b> |
| Cell Viability                 | $\geq 90\% \pm 0.1$ of requested cells                                                                                  | Within range                | <b>QUALITY CONTROL APPROVED</b> |
| Sterility                      | No cloudiness of the test media                                                                                         | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Bacterial Endotoxin Test (LAL) | Endotoxin sensitivity 0.125 EU/mL                                                                                       | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Mycoplasma                     | Mean of absorbance $< 0.05$                                                                                             | Negative                    | <b>QUALITY CONTROL APPROVED</b> |
| Immunophenotyping              | Positive marker – CD73, CD90, CD105 ( $\geq 95\%$ )<br>Negative marker – CD11b, CD19, CD34, CD45, HLA-DR ( $\leq 2\%$ ) | Within range                | <b>QUALITY CONTROL APPROVED</b> |

The cell product delivered by Malaysian Genomics Resource Centre Berhad is manufactured and tested for research purpose only. Product is processed in a facility operating under current Good Manufacturing Practice (cGMP) guidelines in accordance with PIC/S standards.

**Authorised Signature:**

**Quality Control Department**

Powered By:



# Certificate of Analysis

## Supporting Information

### MORPHOLOGY OF CELLS

Figure 1: Mesenchymal stem cells as viewed under an inverted phase contrast microscope (10x).

### STERILITY TEST <sup>6, 8</sup>

Aerobic microorganisms are incubated in Soybean Casein Digest Medium at 20°C - 25°C for 14 days.

Facultative microorganisms are incubated in Fluid Thioglycollate Medium at 30°C - 35°C for 14 days.

### BACTERIAL ENDOTOXIN TEST (LIMULUS AMOEBOCYTE LYSATE, LAL) TEST <sup>3,6</sup>

Negative findings for Gram-negative bacteria in cell culture based on FDA Reference of Standard Endotoxin Guideline.

### MYCOPLASMA TEST <sup>4,6</sup>

Negative findings for mycoplasma species *M. hyorhinis*, *M. arginini*, *M. fermentans*, *M. orale*, *M. Pirum*, *M. hominis*, *M. salivarium* and *A. laidlawii*.

### IMMUNOPHENOTYPING

| Marker     | Positive |       |        | Negative (CD11b, CD19, CD34, CD45, HLA-DR) |
|------------|----------|-------|--------|--------------------------------------------|
|            | CD73+    | CD90+ | CD105+ |                                            |
| Result (%) |          |       |        |                                            |

Table 1: Flow cytometric analysis was performed to verify the characteristic surface marker expression profile of mesenchymal stem cells.

Powered By:



### REFERENCES

1. Dominici, M., Blanc, K. Le, Mueller, I., Slaper-Cortenbach, I., Marini, F.C., Krause, D.S. Deans, R.J., Keating, A., Prockop, D.J., Horwitz, E.M. (2006) Minimal Criteria for Defining Multipotent Mesenchymal Stromal Cells. The International Society for Cellular Therapy Position Statement. Cytotherapy, 8, 4-15-17.  
2. Bacterial Endotoxins. European Pharmacopoeia 5.0; <https://gmp.eu.com/Validation/Method/LAL/ELPHARMACPOEIA.pdf>  
3. US Pharmacopeia <8> Mycoplasma Tests  
4. Bobis, S., Jerocha, O., Majla, M. (2008). Mesenchymal stem Cells: Characteristics and Clinical Applications. Folia Histochemica et Cytobiologica. 44(4): 215-230.  
5. USP Method: Sterilization of Medical Devices, Microbiological Methods.  
6. ICA Harmonized Tripartite Guideline, Specifications: Test Procedures and Acceptance Criteria for Biotechnological / Biological Products Q6B.  
7. Muntón, S., Sánchez-Guajó, F.M., Carrancio, S., Villarán, E., López, O., Díez-Camelo, M., San Miguel, J.F. and del Caizo, M.C. (2012). Optimisation of mesenchymal stromal cells karyotyping analysis: implications for clinical use. Transfusion Medicine, 22: 122-127.  
8. US Pharmacopeia <7> Sterility Test